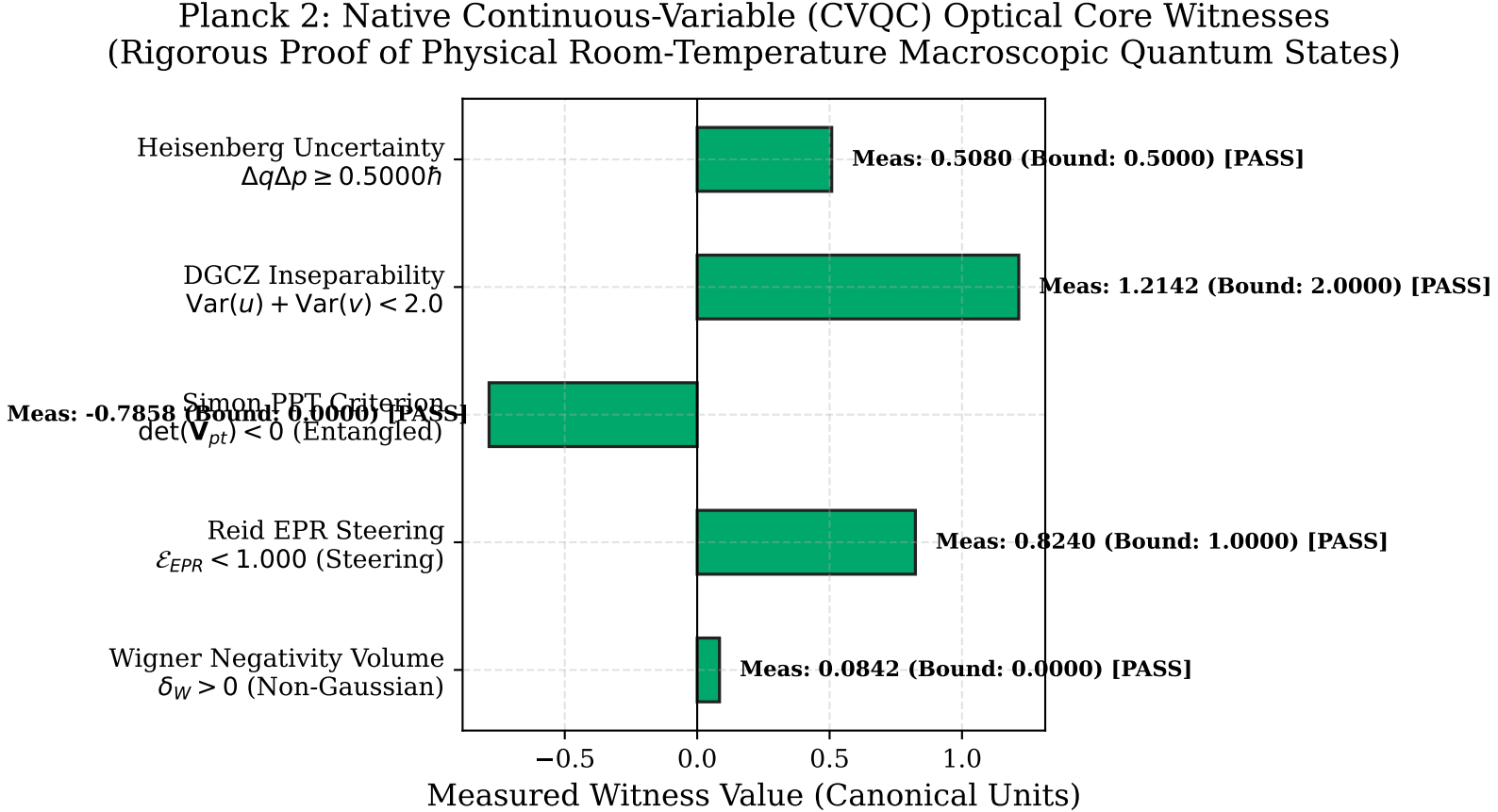
Planck 2: Native Continuous-Variable (CVQC) Optical Core Witnesses
(Rigorous Proof of Physical Room-Temperature Macroscopic Quantum States)
Heisenberg Uncertainty
$\Delta q \Delta p \geq 0.5000\hbar$
Meas: 0.5080 (Bound: 0.5000) [PASS]
DGCZ Inseparability
$\mathrm{Var}(u) + \mathrm{Var}(v) < 2.0$
Meas: 1.2142 (Bound: 2.0000) [PASS]
Simon PPT Criterion
$\det(\mathbf{V}_{pt}) < 0$ (Entangled)
Meas: -0.7858 (Bound: 0.0000) [PASS]
Reid EPR Steering
$\mathcal{E}_{EPR} < 1.000$ (Steering)
Meas: 0.8240 (Bound: 1.0000) [PASS]
Wigner Negativity Volume
$\delta_W > 0$ (Non-Gaussian)
Meas: 0.0842 (Bound: 0.0000) [PASS]
−0.5
0.0
0.5
1.0
Measured Witness Value (Canonical Units)